PhiCor.ai

The Mathematical Human Body

Comprehensive Feature Documentation

1,977
Disease Models

336+
Drug Profiles

18
Organ Systems

10
Biological Scales

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📑 Table of Contents

1. Overview
2. Main Clinical Dashboard
3. Multi-Scale Parameter System
4. Disease Models with Examples
5. Drug Engine
6. Differential Diagnosis
7. Expert Curation Portal
8. Knowledge Bank
9. User Management
10. API Reference

1. Overview

PhiCor.ai is a comprehensive digital twin platform that mathematically models the human body across multiple biological scales. It provides clinicians, researchers, and medical experts with an interactive tool to explore physiological parameters, disease modifications, drug effects, and differential diagnoses.

Key Features

🏥 1,977 Disease Models

AI-enhanced parameter modifications for comprehensive disease representation.

💊 336+ Drug Profiles

Complete physiological effects modeling for common medications.

🫀 18 Organ Systems

Heart, Brain, Lungs, Liver, Kidneys, and 13 more organ systems.

📊 10 Biological Scales

Geometry, Pressures, ECG, Cellular, Lab Tests, and more.

Technology Stack

- **Frontend:** React.js with Tailwind CSS & shadcn/ui
- **Backend:** FastAPI (Python)
- **Database:** MongoDB
- **AI:** OpenAI GPT-4o via Emergent LLM Key

2. Main Clinical Dashboard

The main dashboard provides an interactive interface for exploring physiological parameters across all organ systems. Users can select organs from the body map, view parameters by scale, and apply disease modifications to see how values change.

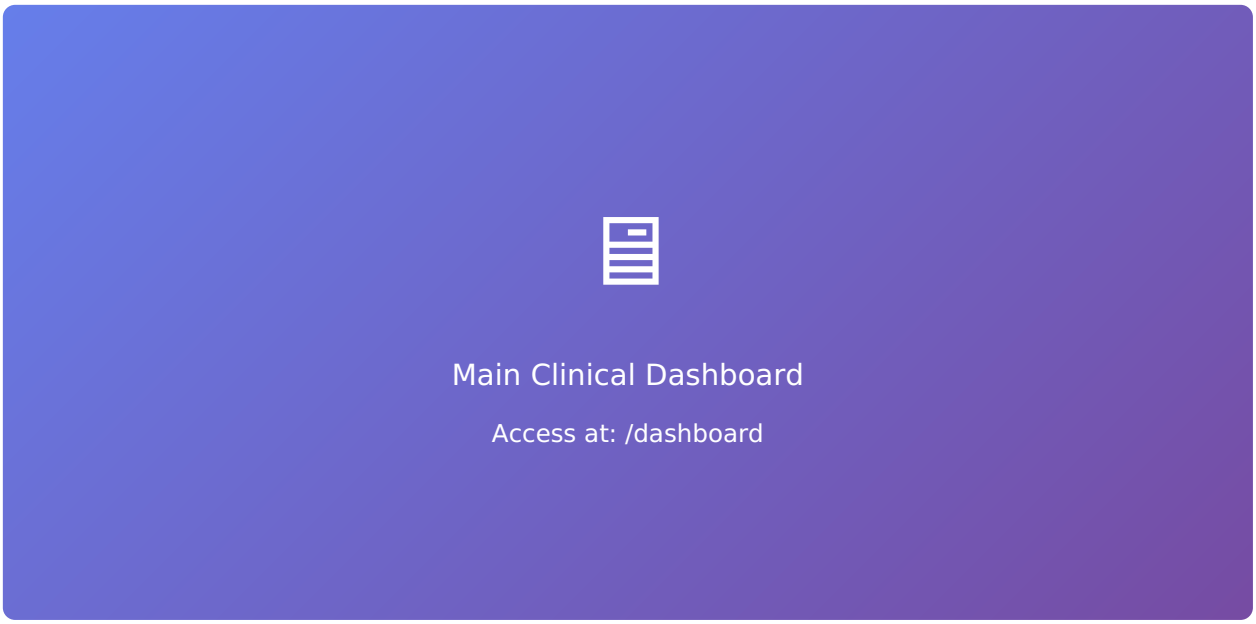


Figure 2.1: Main Clinical Dashboard with body map, parameter grid, and disease search

Dashboard Components

- **Body Map (Left):** Interactive visualization of 18 organ systems
- **Parameter Grid (Center):** Multi-scale parameter display with values and ranges
- **Disease Search (Top):** Search and apply disease modifications
- **Feature Panels (Right):** Quick access to:
  - **ρ (rho)** - Drug Engine
  - **Δ (delta)** - Differential Diagnosis
  - **λ (lambda)** - Clinical Calculator
  - **Ω (omega)** - Timeline

3. Multi-Scale Parameter System

PhiCor organizes physiological data across 10 biological scales, providing a comprehensive view of human physiology from anatomical structures down to cellular processes.

Scale	Description	Example Parameters
📐 Geometry	Physical dimensions	LV wall thickness, LA diameter
🌬 Pressures	Hemodynamic values	LVEDP, PAP, PCWP
🧱 Structure	Tissue composition	Fibrosis %, Calcification
⚡ Physiology	Functional measures	EF, CO, GFR
🔬 Cellular	Molecular level	Ion concentrations, Receptors
🧪 Lab Tests	Blood/urine values	Troponin, BNP, Creatinine
📷 Imaging	Radiological findings	CT/MRI/Echo measurements
📈 ECG	Electrical activity	PR, QRS, QTc intervals
❤ Vitals	Clinical observations	HR, BP, RR, Temp, SpO2
👁 Physical	Exam findings	Murmurs, Edema, JVP

4. Disease Models with Examples

PhiCor contains 1,977 disease models with AI-enhanced parameter modifications. Each disease includes detailed parameter changes across relevant organ systems and scales.

📊 **Data Enhancement Progress**

- **Phase 1 (Basic Enhancement):** ~94% complete (1,859/1,977)
- **Phase 2 (Comprehensive Models):** ~21% complete (424/1,977)

Example: Atrial Fibrillation

When Atrial Fibrillation is applied, the following parameter modifications occur:

Parameter	Normal	Modified	Change
PR Interval	160 ms	ABSENT	No P waves
Heart Rate	72 bpm	100-180 bpm	Irregular
QTc Interval	420 ms	480 ms	+14.3%
Cardiac Output	5 L/min	4.25 L/min	-15%
P Wave	Present	Absent	Fibrillatory waves

Example: Heart Failure (HFrEF)

Parameter	Normal	Modified	Change
Ejection Fraction	55%	30%	-45%
LV Wall Thickness	12 mm	14 mm	+17%
LA Diameter	40 mm	52 mm	+30%
BNP	<100 pg/mL	850 pg/mL	↑↑↑
QRS Duration	90 ms	140 ms	+56%

5. Drug Engine

The Drug Engine contains 336+ drug profiles with detailed physiological effects. Drugs are categorized by therapeutic class and show how they modify body parameters.

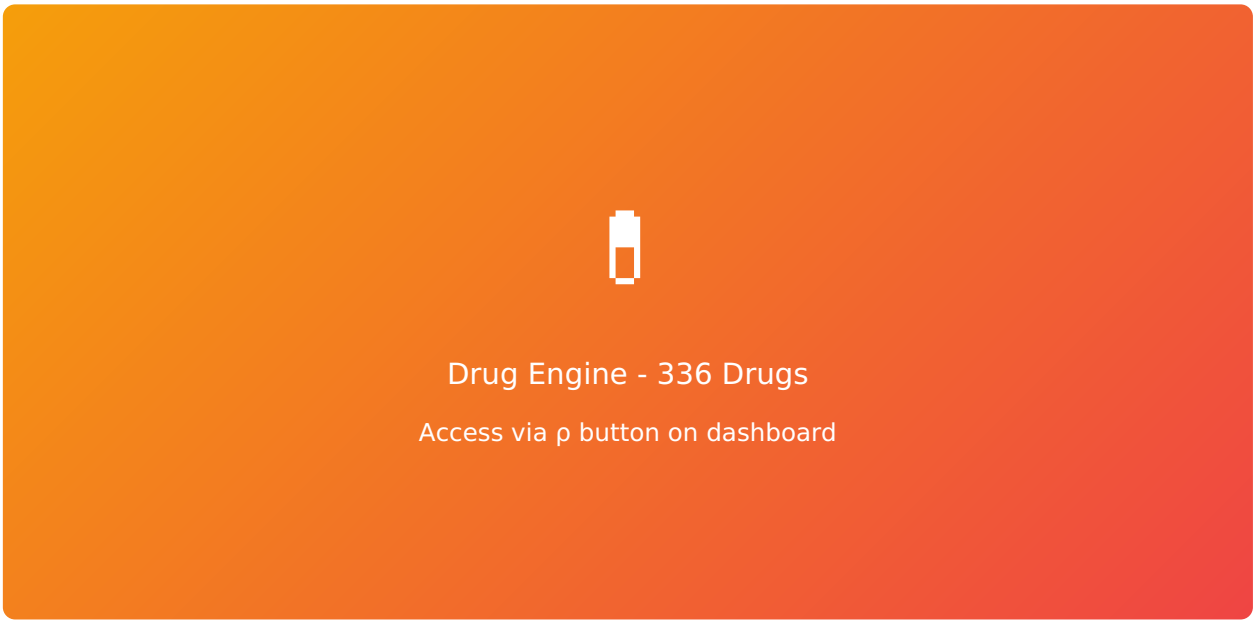


Figure 5.1: Drug Engine with 12 categories and searchable drug profiles

Drug Categories

❤ Cardiovascular

ACE inhibitors, Beta-blockers, Antiarrhythmics, Diuretics

🫁 Respiratory

Bronchodilators, Corticosteroids, Antihistamines

🧠 CNS

Antidepressants, Antipsychotics, Anxiolytics

🦠 Antimicrobial

Antibiotics, Antivirals, Antifungals

Example: Metoprolol

Class: Beta-1 Selective Blocker

Parameter Effects:

- Heart Rate: -20 to -30%
- Blood Pressure: -10 to -15%
- Cardiac Output: -5 to -10%
- AV Conduction: Slowed (PR prolongation)

6. Differential Diagnosis

The AI-powered differential diagnosis panel analyzes abnormal parameters and generates a ranked list of possible diagnoses with probability scores and key distinguishing features.

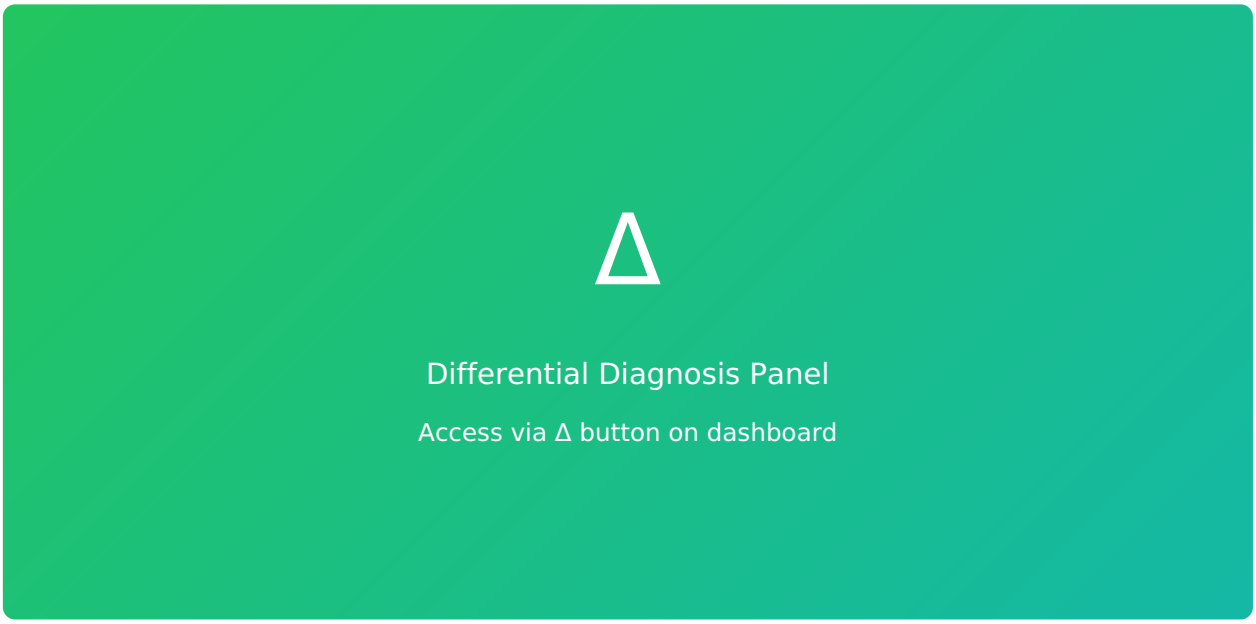


Figure 6.1: Differential Diagnosis with probability scoring and key features

How It Works

1. Select an organ system with abnormal parameters
2. AI analyzes the parameter pattern
3. Generates ranked differential diagnoses
4. Each diagnosis shows: probability score, supporting findings, differentiating features

Example: Elevated Troponin + ST Changes

Diagnosis	Probability	Key Features
Acute Coronary Syndrome	85%	Troponin↑, ST changes, chest pain
Myocarditis	12%	Viral prodrome, diffuse ST changes
Takotsubo Cardiomyopathy	3%	Emotional trigger, apical ballooning

7. Expert Curation Portal

The Expert Curation Portal allows medical specialists to review, verify, and update the platform's data. Experts can suggest changes to parameters, which are then reviewed and approved by administrators.

🔐 **Access Credentials**
URL: /expert-login

Email: admin@phicor.ai

Password: admin123

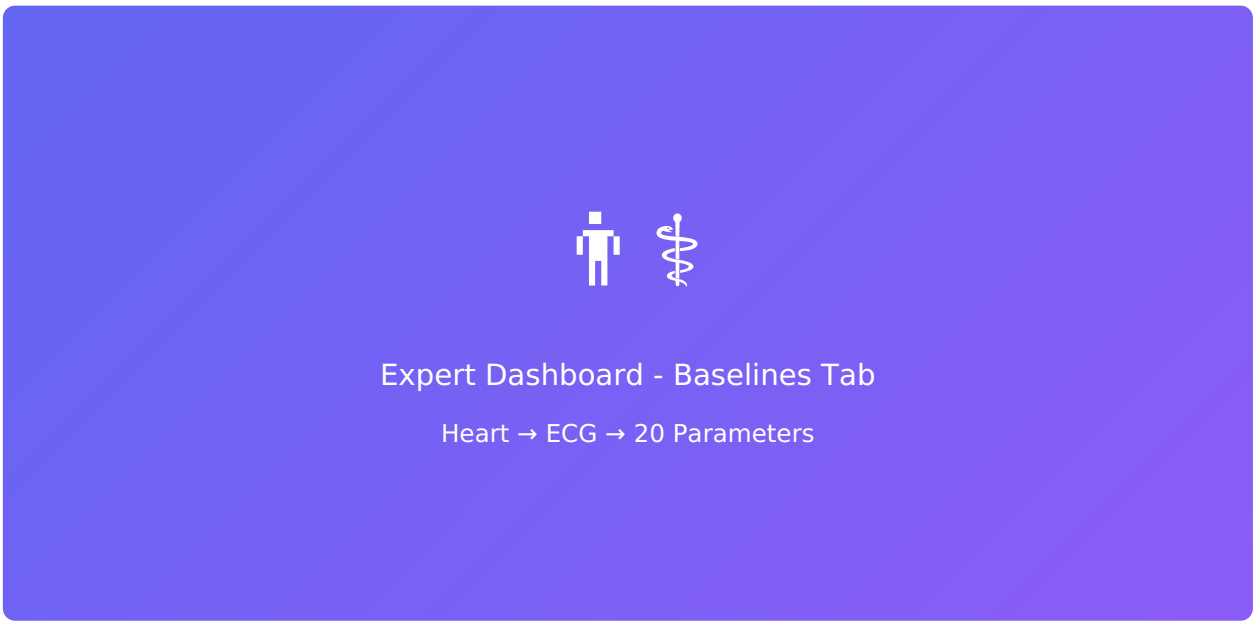


Figure 7.1: Expert Dashboard showing ECG baseline parameters for the Heart

Portal Tabs

Tab	Description
Baselines	Review normal/standard parameters by organ and scale
Diseases	Review disease-specific parameter modifications (1,581 diseases)
References	Lab tests, guidelines, and protocols (396 items)
Changes	Pending change requests for approval
Guidelines	AI-detected guideline updates
Lab	Experiment sandbox for testing

Knowledge (Admin)	Upload documents for AI analysis
Users (Admin)	User management

8. Knowledge Bank

The Knowledge Bank allows administrators to upload medical documents, URLs, and text for AI analysis. The system automatically extracts parameter updates, guideline changes, and clinical modifications, creating suggestions for admin approval.

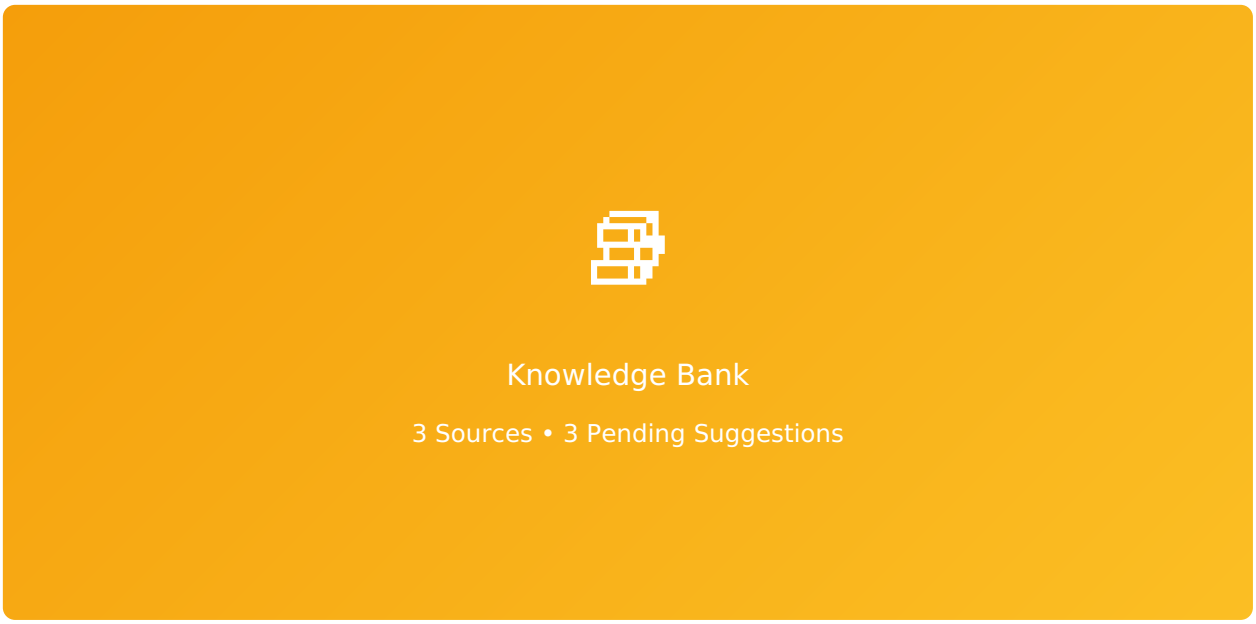


Figure 8.1: Knowledge Bank with AI-analyzed sources and pending suggestions

Features

- **Upload Sources:** URLs, PDFs, Documents, or direct text input
- **AI Analysis:** GPT-4o extracts medical updates automatically
- **Suggestions:** Review proposed changes with evidence levels and confidence scores
- **Approval Workflow:** Approve or reject each suggestion
- **Auto-Search:** Scheduled AI search for new guidelines (configurable 7-90 days)

Workflow

1. Click "Add Knowledge"
2. Select type: URL / Document / Text
3. Enter title and content
4. Optionally link to a disease
5. Click "Upload & Analyze"
6. AI extracts suggestions automatically
7. Review and approve/reject each suggestion

9. User Management

The User Management system allows Super Admins to create, edit, and manage user accounts. Users are assigned roles with different permission levels.

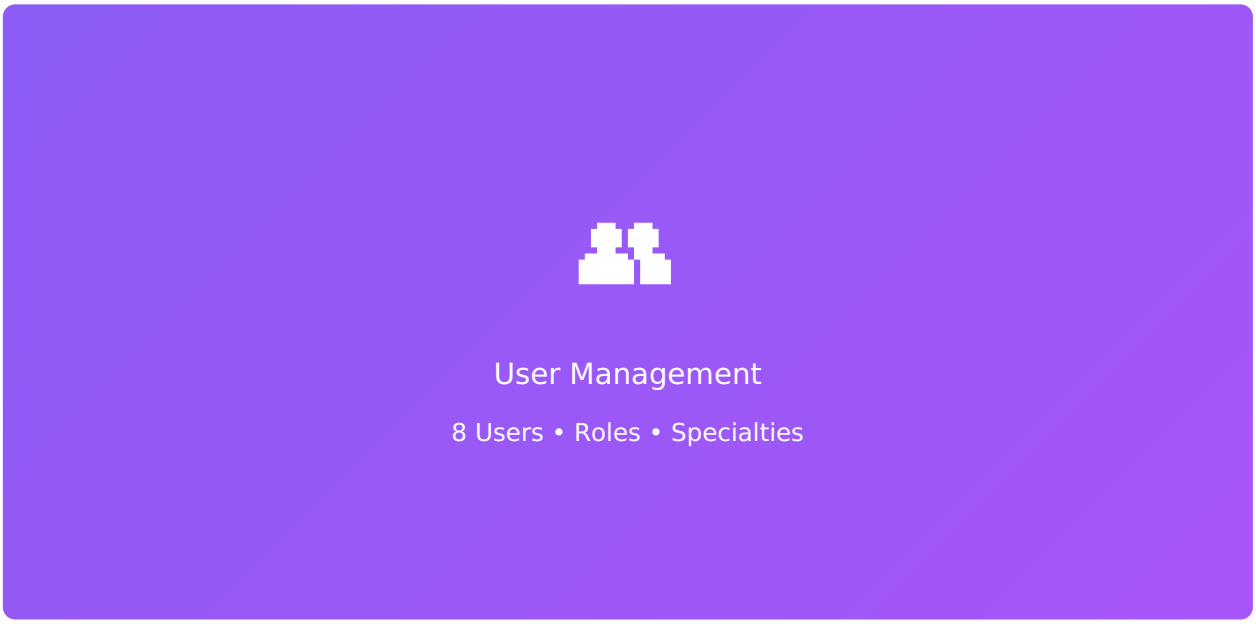


Figure 9.1: User Management showing all users with roles and actions

User Roles

Role	Permissions
Expert	View parameters, suggest changes, run experiments
Admin	All Expert + approve changes, view audit logs
Super Admin	All Admin + manage users, system settings

User Actions

- **Create:** Add new experts with role and specialty
- **Edit:** Update user details, change role
- **Reset Password:** Set new password for user
- **Deactivate:** Soft-delete (can be reactivated)

10. API Reference

PhiCor.ai provides a comprehensive REST API for integration with external systems. All protected endpoints require JWT authentication.

Authentication

```
POST /api/experts/login
Body: {"email": "...", "password": "..."}
Returns: {"access_token": "...", "expert": {...}}
```

All protected endpoints require: `Authorization: Bearer {token}`

Core Endpoints

Endpoint	Method	Description
`/api/experts/login`	POST	Authenticate and get JWT token
`/api/experts/diseases-to-review`	GET	Get all diseases with modifications
`/api/disease/{id}/parameters`	GET	Get parameters for specific disease
`/api/change-requests`	POST	Submit parameter change request
`/api/admin/knowledge/upload`	POST	Upload knowledge source
`/api/admin/knowledge/suggestions`	GET	Get pending AI suggestions
`/api/admin/experts`	GET/POST	List/Create experts
`/api/admin/settings`	GET/PUT	System settings
`/api/enhancement/status`	GET	Check data enhancement progress

Document Information

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Total Diseases: 1,977

Total Drugs: 336+